

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: pGL-luciferase vector [Promega]containing b-globin promoter ; co-porated SV 40, b-gal.
Species Used	Human, K562, chronic myeloid leukemia; HeLa, epithelial carcinoma; HEL cells, erythroleukemia.		

Before the Pulse

Cell Growth Medium	RPMI + 5% Fetal Calf Serum + 5% DCS	Growth Phase at Harvest	2 to 5 x 10 (5) cells / ml
		Pre-pulse Incubation	10 min., ice

Wash Solution Phosphate Bufferd Saline + 5% DCS

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	25°C
Electroporation Medium*	Hepes Buffered Saline

Cuvette Gap 0.4 cm

Cell Density 4 x 10 (7) cells / ml

Voltage 0.300 kV

Volume of Cells 0.5 ml

Field Strength 0.75 kV/cm

DNA Concentration 50 µg per pulse

Capacitor 960 µF

DNA Resuspension Buffer Not given

Resistor (Pulse Controller) Ω none

Volume of DNA 50 µl

Time Constant 31 msec

After the Pulse

Outgrowth Medium RPMI + 5% Fetal Calf Serum + 5% DCS

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	24 hours
Selection Method or Assay Used	luciferease, β-gal

Electroporation Efficiency Not given

Per Cent Survival 50%

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Survey Number

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